

NEUMAYER STATION III, Antarctica

WMO: 890020

Lat: 70.676S

Lon: 8.274W

Elev: 164

StdP: 14.61

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(1) 7	-41.2	-37.9	-46.6	0.4	-40.6	-43.2	0.5	-37.5	65.1	5.6	59.9	0.6	9.1	150	1.472 (1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
		0.4%		1%		2%		0.4%		1%		2%				
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2) 1	9.5	31.8	30.3	30.8	29.4	29.7	28.4	30.7	31.5	29.7	30.6	28.6	29.6	17.5	80	(2)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(3) 29.8	24.0	30.9	28.5	22.6	30.1	27.2	21.3	29.1	11.4	31.7	10.9	30.7	10.5	29.7	33.1 (3)

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 58.9	53.1	47.7	DB	-46.8	33.5	4.3	1.6	-49.9	34.6	-52.4	35.6	-54.9	36.4	-58.0	37.6	(4)
(5)			WB	-46.1	31.7	2.8	1.0	-48.1	32.4	-49.8	32.9	-51.4	33.5	-53.5	34.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	23.5	16.8	7.1	-0.9	-5.0	-10.3	-13.6	-12.1	-10.7	0.1	13.8	22.8		
	DBStd	16.81	4.65	6.81	9.43	12.02	13.74	12.50	12.52	13.11	12.23	10.01	6.84	4.51		
	HDD50	17317	822	930	1330	1526	1706	1808	1973	1926	1820	1548	1085	843		
	HDD65	22792	1287	1350	1795	1976	2171	2258	2438	2391	2270	2013	1535	1308		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	19.9	14.7	17.5	20.1	21.0	22.1	20.4	20.2	22.8	21.5	21.6	20.3	16.6		
	PrecAvg	16.1	1.3	1.3	1.4	1.3	1.6	1.0	1.0	1.3	1.1	1.5	1.9	1.5		
(15) Precipitation	PrecMax	23.1	2.8	2.8	3.5	2.4	2.9	2.8	2.4	2.0	3.0	3.1	4.1	3.4		
	PrecMin	10.7	0.3	0.3	0.4	0.4	0.4	0.3	0.2	0.4	0.3	0.2	0.6	0.4		
	PrecStd	2.7	0.6	0.6	0.7	0.6	0.6	0.6	0.5	0.5	0.6	0.7	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	32.0	27.5	26.0	25.1	19.1	16.7	19.4	19.7	24.3	28.9	33.7		
		MCWB	30.9	30.9	26.8	25.6	24.4	18.5	16.1	18.4	18.9	23.3	27.6	31.2		
	2%	DB	32.0	30.2	24.6	21.7	20.0	15.2	11.8	14.1	12.5	19.6	26.8	31.5		
		MCWB	30.6	29.3	23.8	20.8	19.4	14.7	11.0	13.6	11.5	18.7	25.5	29.5		
	5%	DB	31.2	28.3	22.0	18.0	16.3	10.7	7.5	10.0	9.0	16.3	25.2	30.4		
		MCWB	29.9	27.1	21.1	17.1	15.5	9.9	6.9	9.5	8.3	15.4	24.1	28.7		
	10%	DB	30.3	26.5	19.7	15.1	13.4	7.7	3.9	6.0	5.5	13.6	23.6	29.3		
		MCWB	29.0	25.3	18.6	14.2	12.6	7.1	3.3	5.4	4.8	12.7	22.7	27.8		
	0.4%	WB	32.1	31.3	27.1	25.6	24.5	18.6	16.1	18.8	19.3	23.5	28.3	31.5		
		MCDB	32.4	31.8	27.3	25.9	25.0	19.0	16.5	19.4	19.9	23.8	28.6	32.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.1	29.4	23.9	20.7	19.5	14.7	11.1	13.6	11.6	18.7	25.8	30.1		
		MCDB	31.6	30.0	24.5	21.4	19.8	15.3	11.8	14.1	12.3	19.4	26.5	31.0		
	5%	WB	30.2	27.3	21.2	17.3	15.6	10.1	6.9	9.4	8.3	15.4	24.2	28.9		
		MCDB	31.0	28.1	21.8	18.0	16.3	10.6	7.5	10.0	9.0	16.1	25.0	30.0		
	10%	WB	29.2	25.4	18.7	14.3	12.6	7.1	3.3	5.5	4.9	12.7	22.7	28.0		
		MCDB	30.2	26.3	19.6	15.1	13.3	7.6	3.8	6.0	5.5	13.5	23.5	29.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	11.3	12.0	12.9	13.0	14.2	13.7	12.9	12.9	12.1	11.1	9.2		
		MCDBR	8.2	9.8	10.0	12.4	11.6	13.7	13.2	10.3	12.0	11.1	10.3	8.1		
	5% WB	MCWBR	7.5	9.3	9.9	12.3	11.6	13.6	12.9	10.2	11.9	10.8	9.7	7.2		
		MCWBR	7.1	8.9	9.5	12.3	11.6	13.8	13.1	9.9	12.1	10.7	9.6	7.6		
	5% WB	MCWBR	6.6	8.6	9.5	12.2	11.5	13.7	12.9	9.8	12.0	10.4	9.2	7.0		
		MCWBR	6.6	8.6	9.5	12.2	11.5	13.7	12.9	9.8	12.0	10.4	9.2	7.0		
(40) Clear-Sky Solar Irradiance	taub	0.261	0.248	0.219	0.159	N/A	N/A	N/A	N/A	0.153	0.198	0.221	0.241	0.259		
	taud	2.257	2.297	2.282	2.053	N/A	N/A	N/A	N/A	1.998	2.197	2.211	2.226	2.227		
	Ebn at Noon	310	294	265	187	0	0	0	0	183	269	303	317	316		
	Edh at Noon	35	29	20	10	0	0	0	0	10	21	30	36	38		
(44) All-Sky Solar Radiation	RadAvg	2127	1379	658	170	0	0	0	0	76	487	1212	2068	2478		
	RadStd	134	112	62	17	0	0	0	0	5	33	57	84	136		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page